MIL-B-273E
17 September 1985
SUPERSEDING
MIL-B-273D
27 March 1980

MILITARY SPECIFICATION

BAG, DRINKING WATER STORAGE: DISPENSING, CLOTH, COTTON DUCK, POROUS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a drinking water storage bag fabricated from cotton duck, and tape, with miscellaneous hardware.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

CCC-D-950	-	Dyeing and Aftertreating Processes for Cotton Cloths
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4610

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MIL-P-116 - Preservation-Packaging, Methods of

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-147 - Palletized Unit Loads

DRAWINGS

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

13200E6480 - Bag, Drinking Water Storage: Dispensing, Cloth, Cotton Duck
13200E6481 - Bag Body
13200E6482 - Faucet
13200E6483 - Cover
13200E6484 - Ring-Body

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Guide sample. Samples of the item, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this document may appear in the sample, in which case this document shall govern.

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.4).

3.3 Materials and components. Materials (see 6.5) and components shall be as specified on Drawings (see 2.1).

3.4 Construction. The construction shall conform in all respects to the drawings and as specified herein.

3.4.1 Automatic stitching. Automatic stitching machines may be used to perform any of the stitch patterns provided the requirements for the stitch pattern, stitches per inch, and size and type of thread are met; and at least three tying, overlapping or backstitches are used to secure the ends of stitching.

3.4.1.1 Type 301 stitching. Ends of stitching shall be backstitched or overstitched not less than 1 inch except where ends are turned under or caught in other seams and stitching. Thread tensions shall be maintained so that there will be no excessively tight stitching resulting in puckering of the materials sewn.

3.4.1.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When bobbin run-outs occur during stitching, the stitching shall be repaired by restarting the stitching a minimum of 1 inch (1/2 inch for box-x and W-W stitching) back of the end of the stitching. 1/

b. When two or more consecutive skipped or run-off stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of 1 inch in back of the defective area (1/2 inch on box-x and W-W stitching), continue over the defective area and continue a minimum of 1 inch (1/2 inch on box-x and W-W stitching) beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

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3.4.2 Fusing of ends of nylon tape and cord ends. All exposed raw ends of nylon tape and cord shall be fused. The apparatus used to fuse the tape and cord ends shall be capable of providing sufficient heat to provide a smooth edge with the cut ends of the yarns all fused together.

3.4.3 Setting of grommets. Holes shall be prepunched to receive the grommets. Holes prepunched to receive the grommets shall be smaller than the outside diameter of the grommet barrel so that the barrel must be forced through the hole. The grommet shall be securely clinched without cutting the adjacent material.

3.5 Dyeing and aftertreatment.

3.5.1 Color, water repellency treatment. Cotton duck cloth for the bag body shall be type I, class C in accordance with CCC-D-950, dyed Olive Drab 7. Cotton duck cloth for the cover shall be rendered water repellent in accordance with CCC-D-950, type I, Class A, dyed Olive Drab 7.

3.5.1.1 Resin treatment. Cotton duck cloth for the bag body shall be treated with resin after dyeing by impregnating the cloth with a water-dispersed copolymer methyl and ethyl acrylate resin, modified with a water-soluble cellulose ether. The side of the cloth that will form the inside of the bag shall be coated with the resin material. The treated cloth shall not adhere to itself and shall provide a seepage of water of not less than 0.2 or more than 0.5 milliliter per square centimeter per hour, when tested as specified herein.

3.5.1.2 Leakage. The level of water in the bag after standing for 24 hours shall fall not more than 10 inches or less than 4 inches, when tested as specified herein.

3.6 Marking. The identification marking shall be printed or stenciled in the location shown on the applicable drawing and shall conform to MIL-STD-130.

3.6.1 Location marks. Location marks shall not be drilled. Printed markings for component location shall not be more than 1/32 inch wide.

3.7 Repairs. Repairs such as mends, darns, patches or splices are not permitted on the bag.

3.8 Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.4.3, shall be removed from production and replaced with nondefective and properly matched components.

3.9 Workmanship. The finished and assembled bag shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the applicable acceptable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested for the characteristics specified in 4.4.5. The presence of any defect or failure to pas any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 Certification. The contractor shall furnish a certificate of compliance for the resin treatment specified in 3.5.5.1 and formulations required to evaluate the safety of the material for the intended use.

4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine whether cut lengths, cut parts, markings for location of components, and location of assembled component parts are in accordance with specified requirements, and that the prepunched holes for receiving grommets are as specified in 3.4.3. Whenever nonconformance is noted, corrections shall be made to the parts affected and lot in process. Parts which cannot be corrected shall be removed from production.

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4.4.3 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of bags. The sample unit shall be one bag. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Fabric	Not as specified	X	
	Hole, cut, tear, smash, broken or missing yarn, or open place clearly visible at normal inspection distance (approximately 3 feet)	X	
Tape	Not firmly and tightly woven; edges frayed or scalloped	X	
	Multiple floats		X
	Any cut, hole, tear, or smash	X	
	Abrasion mark, slub, broken end or pick		X
Tape and cord	Ends not fused as specified (where required)		X
Hardware (general)	Broken or malformed failing to serve intended purpose, corroded area, burr, or sharp edge	X	
	Finish omitted, partially omitted, or not as specified on brass		X
	Not assembled as specified (unless otherwise classified herein)	X	
	Not specified type, size, or style	X	
Grommets	Insecurely clinched or clinched excessively tight, cutting fabric	X	
	Washer installed on incorrect side		X
Open seam	1/2 inch or less		X
	More than 1/2 inch	X	
NOTE:	A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped or runoff stitches occur.		

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Raw edge (on edge required to be finished)	More than 1/2 inch in length when securely caught in stitching	X	
NOTE:	Raw edge not securely caught in stitching shall be classified as an open seam.		
Runoff (see open seam)			
Seam and stitch type	Wrong seam or stitch type	X	
Stitch tension	Loose, resulting in a loose bobbin or top thread		X
	Excessively tight, resulting in puckering of material		X
Stitches per inch	One stitch less than minimum specified		X
	Two or more stitches less than minimum specified	X	
	One or more stitches in excess of maximum specified		X
NOTE:	Variation in the number of stitches per inch caused by the operator speeding up the machine and pulling the materials in order to sew over heavy places or heavy seams, or in turning corners shall be classified as follows: (a) Within the minor defect classification - no defect (b) Within the major defect classification - minor defect		
Thread breaks, skipped stitches, or runoffs	Not overstitched as specified		X
NOTE:	Thread breaks or two or more consecutive skipped or runoff stitches not overstitched shall be classified as open seams.		

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Ends of stitching	Not secured as specified		X
Rows of stitching	Any row missing	X	
Component and assembly	Any component part omitted or not as specified or any required operation omitted (unless otherwise classified herein)	X	
	Needle chews	X	
	Any mend, darn, patch, or splice	X	
Binding	Loosely applied but not exposing raw edge of material		X
	Loosely applied exposing raw edge of material	X	
Marking	Omitted, illegible, incorrect, or misplaced		X
Location markings	Drilled	X	
	Printed marking more than 1/32 inch in width		X
Cleanness	Any spot or stain clearly noticeable		X
Thread ends	Not trimmed to 1/4 inch maximum length		

4.4.4 End item dimensional examination. The bag shall be examined for conformance to the dimensions specified on the drawings. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of bags. The sample unit shall be one bag. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 10.0.

4.4.5 End item testing. The end item shall be tested as specified in 4.5 for conformance to the requirements specified in 3.5.1.1 and 3.5.1.2. The lot size shall be expressed in units of bags. The sample size shall be the number of bags indicated by inspection level S-2. Any test failure shall be cause for rejection of the lot.

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4.4.6 Packaging inspection. An examination shall be made to determine that preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged. The lot size shall be the number of containers in the inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of component such as: incomplete closure of container flap, loose strapping, inadequate stapling, or improper taping. Bulged or distorted container.
Content	Weight of shipping container is more than specified.

4.4.7 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

4.5 Tests.

4.5.1 Tackiness. Two specimens not less than 3 inches by 5 inches shall be used for this test. Place the coated sides of the specimens together and apply

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a pressure of not less than 0.7 pound per square inch at not less than140°F for not less than 3 hours. Any adherence between the two specimens shall constitute failure of this test.

4.5.2 Seepage of coated cloth. Three specimens not less than 8 inches by 8 inches shall be used for this test. Determine the seepage of the coated cloth in accordance with FED-STD-191, Method 5516, except that height of water shall be 30 inches, the coated side of the cloth shall be next to the water, and exposure time shall be not less than 60 minutes. Seepage of less than 0.2 $ml/cm^2/hr$ or more than 0.5 $ml/cm^2/hr$ shall constitute failure of this test.

4.5.3 Leakage of bag. Conduct the test at a temperature between 70°F and 80°F, and a relative humidity of not more than 65 percent. Suspend the bag by the suspension cords. Fill the bag with water to the bottom of the overflow outlet, and tie the cover in place. After 4 hours, add water as required to bring the level of the water to the bottom of the overflow outlet. The bag shall then remain undisturbed for 24 hours, plus or minus 10 minutes. The water level of any bag dropping more than 10 inches or less than 4 inches below the bottom of the overflow outlet shall constitute failure of this test.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each complete bag shall be compactly folded and preserved in accordance with MIL-P-116, method III in a close-fitting box conforming to PPP-B-636, W6c, style optional. The box shall be waterproofed using method V closure in accordance with the appendix of the box specification.

5.1.2 Commercial. The bags shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Bags preserved as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to overseas type, style optional, grade B of PPP-B-601 or class 2, style optional of PPP-B-621. The gross weight of each shipping container shall not exceed 200 pounds. Closure and strapping shall be in accordance with the applicable box specification.

5.2.2 Level B packing. Bags preserved as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to style RSC, grade V3c or V13 c of PPP-B-636. The weight of contents of each shipping container shall not exceed the weight limitations of the container specification.

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5.2.3 Commercial packing. Bags preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), bags, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type I or Ia of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L, or film bonding means O or P. The pallet pattern shall be No. 32 in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The water storage bags are intended for use in the field as a container for drinking water.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.2, 4.3, and 6.4).
c. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
d. When palletization is required (see 5.3).

6.3 Samples. For access to samples, address the contracting activity issuing the invitation for bids.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample consisting of one finished bag. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.3).

6.6 Leakage. The resinous finish of the bag allows the seepage of a small quantity of water through to the outside surface of the bag where it is evaporated and lowers the temperature of the water in the bag. Water purification chemicals, as prescribed, may be added to the water in the bag to keep the water potable.

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6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - YD
Air Force - 99

Review activities:

Army - MD
Air Force - 82
DLA - CS

User activity:

Navy - MC

Preparing activity:

Army - GL

Project No. 4610-0107

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL–B–273E	BAG, DRINKING WATER STORAGE: DISPENSING, CLOTH, COTTON DUCK, POROUS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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